

1. Incorrectly associated?

- a. Erythema Multiforme (EM) with *Mycoplasma pneumoniae* and herpes simplex
- b. Erysipelas with *S. pyogenes*
- c. Merkel cell carcinoma (MCC) with Polyoma virus
- d. Subacute sclerosing panencephalitis (SSPE) with mulluscum contagiosum virus.
- e. Lichen planus (LP) and Porphyria cutanea tarda with hepatitis C.

2. Skeletal muscle metabolism during exercise:

- a. causes a sharp rise in blood ketone body concentration.
- b. uses only phosphocreatine in the anaerobic state.
- c. shifts from primarily glucose to primarily fatty acids as fuel during aerobic exercise.
- d. uses largely glycogen and phosphocreatine in the aerobic state.
- e. is the same in both aerobic and anaerobic exercise.

3. Which of the following bacteria is the common pathogen found in otitis media?

- a. Viridans streptococcus
- b. *Streptococcus pneumoniae*
- c. *Streptococcus agalactiae*
- d. *Staphylococcus saprophyticus*
- e. *Streptococcus pyogenes*

4. Which one of the following is the correct procedural order in preparing culture media?

- 1. Incubation at ambient temperature,
- 2. Inoculation of specimen,
- 3. Sterilization,
- 4. Weighing and dissolving ingredients in distilled/ deionized water,
- 5. pH checking and adjustment

- a. 1,2,3,4,5

b. 2,3,4,5,1

c. 4,5,3,2,1

d. 3,4,2,1,5

6. Wrong statement about the skeletal muscle

a. Increased neuromuscular coordination produces greater force of contraction

b. Skeletal muscle fibers that rely mainly on glycolysis are fatigue resistant

c. Can be tetanized

d. Acetylcholine is the only neurotransmitter which can excite the skeletal muscle

e. Skeletal muscles that rely mainly on oxidative phosphorylation have high capillary density

7. Which one of the following fungal diseases was responsible for the recent unprecedented outbreak of mycosis amongst COVID-19 patient in India?

a. Mucormycosis

b. Pneumocystosis

c. Cryptococcosis

d. Candidiasis

e. Aspergillosis

8. "Spaghetti and meatballs" appearance on microscopy of a 10% KOH treated specimen is a characteristic of the fungus causing:

a. Candidiasis

b. Tinea versicolor

c. Tinea nigra palmaris

d. Aspergillosis

e. Cryptococcosis

9. Which systemic complication of bacterial skin infection is caused mainly by *S. aureus*?

a. Abscess

b. Sepsis

c. Acute poststreptococcal glomerulonephritis (APSGN)

d. Staphylococcal scalded-skin syndrome (SSSS)

e. A and D

10. A case of leprosy with good Cell mediated immunity usually presents as the following form of disease:

a. Lepromatous

b. Multibacillary leprosy

c. Borderline

d. indeterminate

e. Tuberculoid

11. All of the following laboratory features are the characteristics of *S. saprophyticus* except

a. the bacteria is unable to utilize mannitol

b. Sensitive to Novobiocin

c. growth on MSA

d. negative reaction during slide coagulation test

e. gamma hemolytic reaction on blood agar plate

12. If a 5-year-old child presents with pinkish red color of his cheeks ("slapped cheek" appearance), what type of exanthem is depicted in the situation?

a. Exanthem infectiosum

b. Hand-foot-mouth disease

c. Exanthem subitum

d. Measles

e. Varicella

13.

Which one of the following laboratory methods is used to differentiate *S. pyogenes* from other beta hemolytic gram positive streptococci

a. Bile solubility

- b. PYR test
- c. Production of catalase
- d. CO₂ requirement
- e. CAMP test

14. All of the following statements are correct about the skeletal muscle, Except

- a. Produce tetanic contraction with repeated stimulation
- b. Produces more work when the muscle contracts isometrically than when the muscle contracts isotonicly
- c. Doesn't exhibit contraction in the absence of external stimuli
- d. Sarcoplasmic reticulum is the main source of Ca²⁺ for contraction
- e. Denervation of the muscle results in atrophy

15. Gram staining can help in the quick identification of presence or absence of fungi in a specimen. Which characteristics of a fungus help us to do so?

- a. It is hyphae only that stains poorly gram positive
- b. Both yeasts and hyphae stain poorly gram negative
- c. Yeasts stain poorly gram negative and are 2 to 3 times bigger than bacterial cocci
- d. If specimen has hyphae, it should measure 2 to 3 times wider than bacterial bacilli

16. In a normal healthy skeletal muscle, which of the following events directly occurs as a result of propagation of an action potential to the presynaptic membrane of α -motor neuron?

- a. Opening of voltage gated Na⁺ channels on the NMJ
- b. Depolarization of the T tubule membrane
- c. Opening of voltage-gated Ca²⁺ channels in the presynaptic motor neuron.
- d. Increase in intracellular Na⁺ concentration in the motor neuron terminal
- e. Contraction of the muscle

17. Which of the following is most likely associated with rigor mortis

- a. Digestion of muscle proteins by lysosomes

- b. Rapid sequestration of Ca^{2+} back to the SR
- c. Damage to actin and myosin filaments
- d. Increased myosin ATPase activity

e. Depletion of ATP

18. One is true about KOH

- a. Have no place in specimens containing keratin
- b. It is always used alone

c. It is a wet mount preparation

d. Never dissolves debris

19. All of the following concerning Aspergillosis are correct, Except?

a. Can easily be diagnosed by detecting branching hyphae microscopically

- b. Infections with Aspergillus are usually associated with contaminated grains
- c. Untreated invasive aspergillosis can have mortality approaching 100%
- d. Aspergillus flavus produces a carcinogenic aflatoxin
- e. The primary route of infection is through the respiratory tract

20. The leading etiologic agent for acute osteomyelitis is

- a. Streptococcus pneumoniae
- b. Neisseria gonorrhoeae
- c. Pseudomonas aeruginosa
- d. Staphylococcus aureus

e. Staphylococcus epidermidis

21. The contractile protein of skeletal muscle involving ATPase activity is:

- a. Tropomyosin
- b. Troponin
- c. Actin
- d. Dystrophin

e. Myosin

23. Which of the following statement is true about the contractile response of skeletal muscle

- a. During isometric contraction, the tension goes up and the muscle shortens
- b. Decreased concentration of Ca^{2+} in the ECF highly affects the contractile response of the skeletal muscle.
- c. Contractile response can vary by different hormones

d. Excitation of the Dihydropyridine receptor results in the influx of Ca^{2+} from the ECF

e. Skeletal muscle cells are normally held at a muscle length that maximizes their ability to generate greater force

24. Which of the following finding make you favor the diagnosis of allergic contact dermatitis (ACD) over irritant contact dermatitis (ICD)?

a. Is a non-immunologic reaction.

b. Occurs only in sensitized individual.

- c. Is a Type I (Ig E mediated) hypersensitivity reaction.
- d. Acute symptom starts as stinging and smarting followed by itching
- e. Depends on the concentration of the agent.

25. Child present with maculopapular rash start from the hair line and go down. What is the likely diagnosis?

a. Chickenpox and shingles.

b. Measles.

- c. Pityriasis Versicolor (PV).
- d. Verruca Vulgaris.
- e. Leishmaniasis.

26. With regard to the muscle tissue, which statement is false?

a. Skeletal and cardiac muscle cells possess sarcomeres

b. Smooth muscles do not have myofibrils

c. Sarcolemma is the plasma membrane of a muscle cell

d. Intercalated discs of cardiac muscle correspond to some of the Z lines

e. Smooth muscle cells have centrally placed nuclei

27.False about Atopic dermatitis?

a. Associated with a personal or family history of AD, allergic rhinitis, and asthma.

b. Is a Type IV cell-mediated hypersensitivity reaction.

c. Moves to the flexor creases as the child grows older.

d. Mutations in the filaggrin gene (FLG) is seen in 50% of severe AD.

e. In neonates affect the cheeks, trunk and extensor surfaces.

28.Ringworm of the groin and associated areas is referred to as:

a. Tinea cruris

b. Tinea corporis

c. Tinea barbae

d. Tinea pedis

e. Tinea capitis

29.Genus Staphylococcus are all, but not

a. Capsulated, non-motile and non-spore forming organism

b. Metabolically active

c. Halophiles grow best aerobically

d. All of the members are equipped with an enzyme that can degrade hydrogen peroxide

e. Detection of DNase production could be used as alternative laboratory test to differentiate the species of the genus

30.Which one of the following culture media is best for performing antimicrobial susceptibility testing for gram positive streptococci species

a. Muller Hinton agar with 5% sheep blood

b. Blood agar plate

c. Mannitol Salt agar

d. Tryptic Soya agar with 5% sheep blood

e. Muller Hinton agar

31.If a patient presents with a cough, conjunctivitis, coryza and photophobia along with Koplik spots on his buccal mucosa, what illness do you suspect?

a. Erythema Infectiosum

b. Herpes simplex

c. Measeles

d. Varicela-Zooster

e. Rubella

32.Against which of the following viral infections is giving vaccination to women in their first trimester ill-advised?

a. Roseola Infantum (sixth Disease)

b. Influenza Virus

c. Human parvovirus B19

d. Hepatitis B Virus

e. German measles

33.Which of the following bacterial metabolic product is responsible in causing Staphylococcal scalded skin syndrome (SSSS)?

a. Toxic shock syndrome toxin-1

b. Exfoliatin toxin

c. Leucocidin

d. Coagulase

e. Alpha toxin

34.Granulomatous (non pyogenic) osteomyelitis may be caused by all except

a. Staphylococcus aureus

b. Parasites (hydatid's disease)

c. Syphilis

d. *Mycobacterium tuberculosis*

e. Fungal agents

35. CAMP test is positive used to differentiate _____ from other beta-hemolytic spp showing the classical “arrow-shaped hemolysis” near the staphylococcus streak

a. Viridans streptococci

b. *Streptococcus pneumoniae*

c. *Streptococcus agalactiae*

d. *Enterococcus faecalis*

e. *Streptococcus pyogenes*

36. Describe a Brodie's Abscess

a. It is a septic joint infection

b. sub-acute form of osteomyelitis, presenting as a collection of pus in bone, often with an insidious onset

c. Is kidney infection

d. It is skin abscess

e. Is acute abscess of the skin

37. What serious complications can a measles virus infection lead to?

a. Permanent skin scar

b. A fetal mortality rate between 2-6.5% due to hydrops fetalis

c. multiple congenital defects

d. Subacute sclerosing panencephalitis (SSPE)

e. Testicular pain

38. Excitation–contraction coupling in skeletal muscle involves all of the following events, EXCEPT

a. Conformational change in Dihydropyridine receptor

b. ATP hydrolysis by myosin ATPase

c. Increased conductance of sarcolemma to Na⁺

d. Depolarization of the T-tubule of the muscle membrane

e. Binding of Ca^{2+} to Calmodulin

39. All of the following statements about Acute hematogenous osteomyelitis are correct except

- a. Usually seen in children
- b. Infection generally involves the metaphyses of rapidly growing long bones

c. Is more common in females than males

- d. Staphylococcus aureus is the commonest cause of infection
- e. Is the most common type of bone infection

40. The purpose of the Cori cycle is to shift the metabolic burden in which of the following directions?

- a. cardiac to skeletal muscle
- b. liver to skeletal muscle

c. skeletal muscle to liver

- d. liver to brain
- e. brain to liver

41. The sarcoplasmic reticulum of skeletal muscle cells are made by:

a. Smooth endoplasmic reticulum

- b. Golgi apparatus
- c. Polyribosomes
- d. Rough endoplasmic reticulum
- e. Mitochondria

42. The latch-bridge mechanism in smooth muscle is responsible for

- a. Rhythmic contraction
- b. Fast muscle twitch
- c. Unstable membrane potential
- d. Excitation-contraction coupling

e. Sustained muscle contraction

43. All of the following statements are correct, EXCEPT

- a. Damage to lower motor neurons results in flaccid paralysis

- b. The strength of cardiac muscle contraction depends to a great extent on the concentration of Ca^{2+} in the ECF
- c. Skeletal muscle contraction is caused almost entirely by Ca^{2+} released from the sarcoplasmic reticulum inside the skeletal muscle fiber itself
- d. Damage to upper motor neurons results in flaccid paralysis followed by spastic paralysis in a few weeks
- e. Myasthenia gravis is caused by an autoimmune destruction of voltage gated Ca^{2+} channels on the presynaptic motor nerve.

44. Let's assume we prepared a wet mount preparation from contaminated food like "Injera", which of the following will be incorrect while examining the preparation under the microscope?

- a. It is possible to visualize spores
- b. Aspergillus is the well known medically important mould to be expected
- c. We can see yeast cells
- d. Mycelium can be seen from the mold growth

45. Type I fibers have which of the following properties?

- a. slow twitch, glycolytic, myoglobin rich
- b. fast twitch, glycolytic, myoglobin poor
- c. fast twitch, oxidative, myoglobin poor
- d. fast twitch, oxidative, myoglobin rich
- e. slow twitch, oxidative, myoglobin rich

46. Regarding gas gangrene one of the following is correct

- a. It is due to Clostridium Botulinum infection
- b. It involves widespread necrosis of the muscles but avoids soft-tissues
- c. The clinical features are due to the release of endotoxin
- d. Gas is invariably present in the muscle compartments
- e. Clostridial species are gram—negative spore forming anaerobe

47. Which of the following condition is false for tissue invasion by C. perfringens?

- a. release of toxins

- b. production of large amounts of hydrogen gas
- c. extremely painful tissue swelling at the site of infection
- d. anaerobic condition of the wound

e. production of small amounts of hydrogen gas

48. Oral thrush in immunocompromised people is mainly caused by:

a. *C. albicans*

- b. *A. fumigatus*
- c. *C. neoformans*
- d. *Pneumocystis jiroveci*
- e. Mucorales

49. Scalded skin syndrome is caused by exotoxin (exfoliatins) produced by

a. *Staphylococcus aureus*

- b. *Pseudomonas aeruginosa*
- c. *Streptococcus pneumoniae*
- d. *Streptococcus pyogenes*
- e. *Staphylococcus epidermidis*

50. Which whitish tongue lesion in immunodeficient patient is easily scraped by spatula?

- a. Erythroplakia
- b. Irritation fibroma
- c. Leukoplakia
- d. Candidiasis

e. Hairy leukoplakia

51. During examination of clinical specimens from wound site one of the following group of contaminants are commonly isolated

a. anaerobic gram positive rods

- b. Beta-hemolytic gram negative cocci
- c. CoNS

d. enteric bacteria

e. *S. pneumoniae*